

Secure Programming Sample Test II

Name and Surname: _____

1. Why should passwords not be stored in plain text?
 - A. It increases CPU usage
 - B. It makes login slower
 - C. A database leak would expose all passwords
 - D. Passwords cannot be searched
2. What is the main purpose of a cryptographic hash function?
 - A. Compress files
 - B. Encrypt network traffic
 - C. Produce a fixed-size one-way representation of data
 - D. Generate random passwords
3. What is a collision in hashing?
 - A. A failed login attempt
 - B. A broken database connection
 - C. Two different inputs producing the same hash
 - D. A password reset
4. Why is a salt added before hashing passwords?
 - A. To shorten passwords
 - B. To speed up hashing
 - C. To make identical passwords produce different hashes
 - D. To encrypt the database
5. Which method is generally recommended for password storage?
 - A. Plain text
 - B. Simple custom hash
 - C. SHA-1
 - D. PBKDF2
6. What is SQL Injection?
 - A. A database backup method
 - B. A way to compress SQL queries
 - C. A vulnerability where user input becomes SQL code
 - D. A network protocol
7. What is dangerous about building SQL queries using string concatenation?
 - A. Queries become shorter
 - B. Databases stop working
 - C. Attackers may inject SQL commands
 - D. Data is automatically deleted
8. What is the effect of the input ' OR '1'='1 in an unsafe login form?
 - A. It encrypts the password
 - B. It creates a new user
 - C. It may bypass authentication
 - D. It restarts the database
9. Which approach is considered the safest for database queries?
 - A. String concatenation
 - B. Input filtering only
 - C. Parameterized queries
 - D. Randomized queries
10. What is information leakage?
 - A. Data compression
 - B. Automatic backup creation
 - C. Exposure of sensitive internal information
 - D. File encryption

11. Why is the following message insecure?
`Wrong password: expected secret`
- A. It is too short
 - B. It causes a syntax error
 - C. It reveals sensitive information
 - D. It prevents login
12. Which exception is raised when converting invalid text to an integer?
- A. FileNotFoundError
 - B. ValueError
 - C. ZeroDivisionError
 - D. TypeError
13. Why can `print(e)` inside exception handlers be dangerous?
- A. It deletes logs
 - B. It slows execution
 - C. It may reveal internal system details
 - D. It blocks input
14. What should never appear in application logs?
- A. Login failures
 - B. Error timestamps
 - C. User passwords
 - D. Event counters
15. What is Cross-Site Scripting (XSS)?
- A. A database attack
 - B. A password hashing method
 - C. Injection of malicious scripts into webpages
 - D. A network scan
16. Which type of XSS stores malicious code in a database?
- A. Reflected XSS
 - B. Stored XSS
 - C. Session XSS
 - D. Local XSS
17. What is the purpose of `html.escape()`?
- A. Execute JavaScript safely
 - B. Encrypt HTML files
 - C. Treat user input as text rather than HTML
 - D. Validate passwords
18. Why are sessions needed in web applications?
- A. To speed up browsers
 - B. To compress webpages
 - C. To remember users between HTTP requests
 - D. To encrypt databases
19. What is a session ID in the context of web applications?
- A. An unique string of characters assigned by the server
 - B. Encrypted password
 - C. The IP address of the server
 - D. A verification code used in two-factor authentication
20. What is session hijacking?
- A. Creating a new account automatically
 - B. Changing database permissions
 - C. Stealing a valid session identifier to impersonate a user
 - D. Blocking HTTP requests